

Jared Brooks, PhD

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WORK EXPERIENCE

Abodemine - Property intelligence evolved

Nov. 2025 – Present

Staff Data Scientist

Denver, CO

- Designed and implemented a scalable **entity resolution system** for real estate transaction data (loans, assignments, releases, pre-foreclosures), enabling accurate linking of fragmented records across disparate data sources and significantly improving dataset integrity for downstream analytics
- Architected an LLM-driven **column matching** system that maps ambiguous client loan tape fields to a canonical dataset of 1,000+ attributes using multi-stage candidate generation, semantic validation, deterministic business rules, and human-in-the-loop feedback for scalable schema alignment
- Built a **human-in-the-loop feedback pipeline** capturing client corrections to continuously retrain and improve matching models, increasing system accuracy and adaptability over time
- Developed robust **model monitoring and performance tracking infrastructure**, ensuring reliability, drift detection, and maintainability across production ML systems

Prove - A leading digital identity and Authentication Solutions provider

Jan. 2021 – Nov. 2025

Senior Data Scientist | June 2024 – Present

Denver, CO

- Architected and deployed International TrustScore **ML model** from concept to production, establishing a new **model microservice** framework that accelerated deployment cycles by 200% and expanded product capabilities globally
- Overhauled legacy Assurance Level (AL) verification system by integrating **ML enhancements** that increased AL=3 **accuracy to 99%**, directly strengthening the company's flagship Verified Users product
- Led software engineering advancement initiatives, including developing the team's core **Python repository** with pytest and **GitLab CICD** pipelines that reduced scripting overhead by 70%, and collaborating with Platform Engineering to implement LazerDome, a centralized **AWS** environment that streamlined secure access and enhanced team collaboration

Data Scientist | October 2021 – June 2024

Denver, CO

- Designed and built the Assurance Level identity verification system from concept to production, creating a session-based transaction framework that achieved **90% accuracy** at **20% population coverage**, meeting critical product requirements
- Established comprehensive **model governance** protocols for **tier 1 banking** clients, ensuring regulatory compliance for identity verification systems while maintaining technical documentation and validation frameworks required by financial institutions

EDUCATION

UC Santa Barbara

Sept, 2017

PhD and Masters Degree in Theoretical Physics

Santa Barbara, CA

University of Southern California

May, 2011

BSc in Physics, Minor in Mathematics

Los Angeles, CA

SKILLS, TECHNOLOGIES, & INTERESTS

- **Skills:** Machine Learning; Data Analysis; Python; SQL; Model Governance; APIs; Mentorship; Statistics; R&D Collaboration; Fraud Detection; Feature Engineering
- **Technologies:** Git; AWS; Sagemaker; Looker
- **Interests:** Musical Performance; Weightlifting; Backpacking; Archery; Cooking